

6B2

Q1

[illegible]

```
haider@ubuntu: ~/Desktop/lab3
File Edit View Terminal Help
I am a Parent Process
I am a Parent Process
I am a Parent Process
haider@ubuntu:~/Desktop/lab3$ I am a Child Process
I am a Child Process
I am a Child Process
I am a Child Process
I am a Child Process
I am a Child Process
I am a Child Process
I am a Child Process
I am a Child Process
I am a Child Process
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I am a Child Process
I am a Child Process
I am a Child Process
I am a Child Process
I am a Child Process
I am a Child Process
I am a Child Process
I am a Child Process
```

Q2

```
q1.cpp
1 #include<iostream>
2 #include<unistd.h>
3 #include<sys/wait.h>
4 using namespace std;
5
6 int main(){
7     pid_t pid = fork();
8     if(pid==0){
9         cout<<"Child Process: Listening current directory contents"<<endl;
10        execlp("ls","ls",NULL);
11    }
12    else if(pid>0){
13        wait(NULL);
14        cout<<"Parent Process: Child has finished execution"<<endl;
15    }
16    return 0;
17 }
```

```
haider@ubuntu: ~/Desktop/lab3
File Edit View Terminal Help
haider@ubuntu:~/Desktop/lab3$ g++ q1.cpp -o q1
q1.cpp: In function 'int main()':
q1.cpp:13: error: expected '}' before 'else'
haider@ubuntu:~/Desktop/lab3$ g++ q1.cpp -o q1
haider@ubuntu:~/Desktop/lab3$ ./q1
Child Process: Listening current directory contents
q1 q1.cpp
Parent Process: Child has finished execution
haider@ubuntu:~/Desktop/lab3$
```

Q3

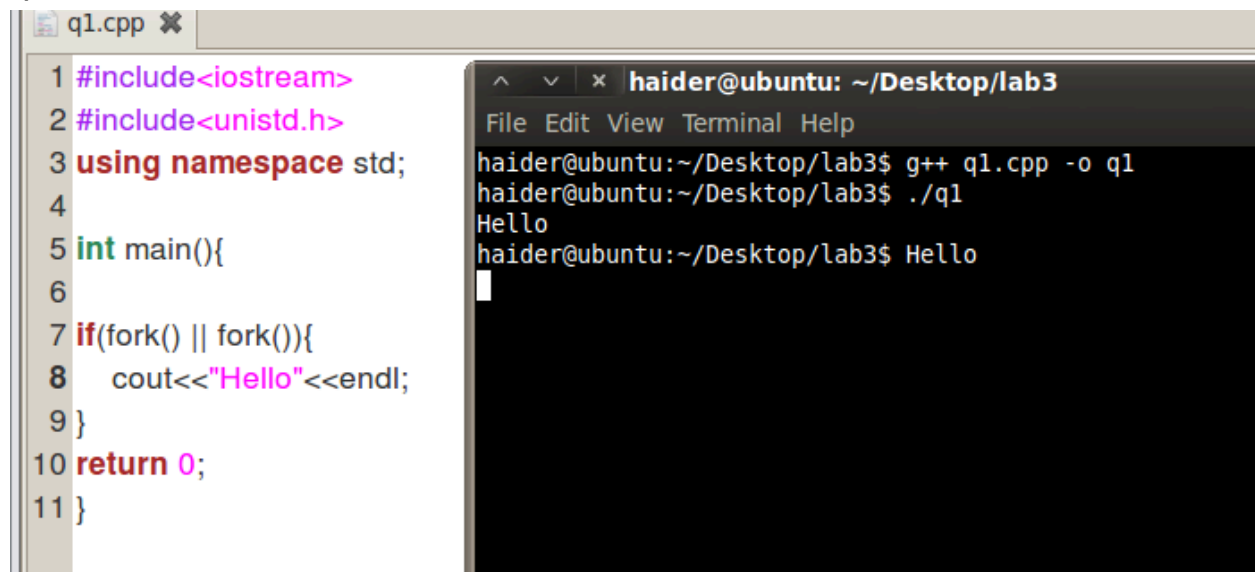
```
q1.cpp ✕
1 #include<iostream>
2 #include<unistd.h>
3 #include<sys/wait.h>
4 #include<stdlib.h>
5 using namespace std;
6
7 int main(int argc, char* argv[]){
8     if(argc<2){
9         cout<<"Usage num1,2,3 "<<endl;
10        return 1;
11    }
12
13    int n = argc-1; //n is size
14    int arr[n];
15    for(int i=0;i<n;i++){
16        arr[i]=atoi(argv[i+1]);
17    }
18    pid_t pid1, pid2, pid3;
19
20    pid1 = fork();
21    if(pid1==0){
22        int sum=0;
23        for(int i=0;i<n;i++){
24            sum+=arr[i];
25        }
26        cout<<"Child1: (Sum)"<< sum<<endl;
27        return 0;
28    }
```

```
haider@ubuntu: ~/Desktop/lab3
File Edit View Terminal Help
haider@ubuntu:~/Desktop/lab3$ g++ q1.cpp -o q1
haider@ubuntu:~/Desktop/lab3$ ./q1 10 5 3 4 2
Child3: (maximum)10
Child2: Average: 4
Child1: (Sum)24
haider@ubuntu:~/Desktop/lab3$
```

```
29 pid2 = fork();
30 if(pid2==0){
31     int sum=0;
32     for(int i=0;i<n;i++){
33         sum+=arr[i];
34     }
35     int avg=sum/n;
36     cout<<"Child2: Average: "<< avg<<endl;
37     return 0;
38 }
39 pid3 = fork();
40 if(pid3==0){
41     int max=arr[0];
42     for(int i=1;i<n;i++){
43         if(arr[i]>max)
44             max=arr[i];
45     }
46     cout<<"Child3: (maximum)"<< max <<endl;
47     return 0;
48 }
49 //parent wait for children
50 wait(NULL);
51 wait(NULL);
52 wait(NULL);
53 return 0;
54 }
```

Q6

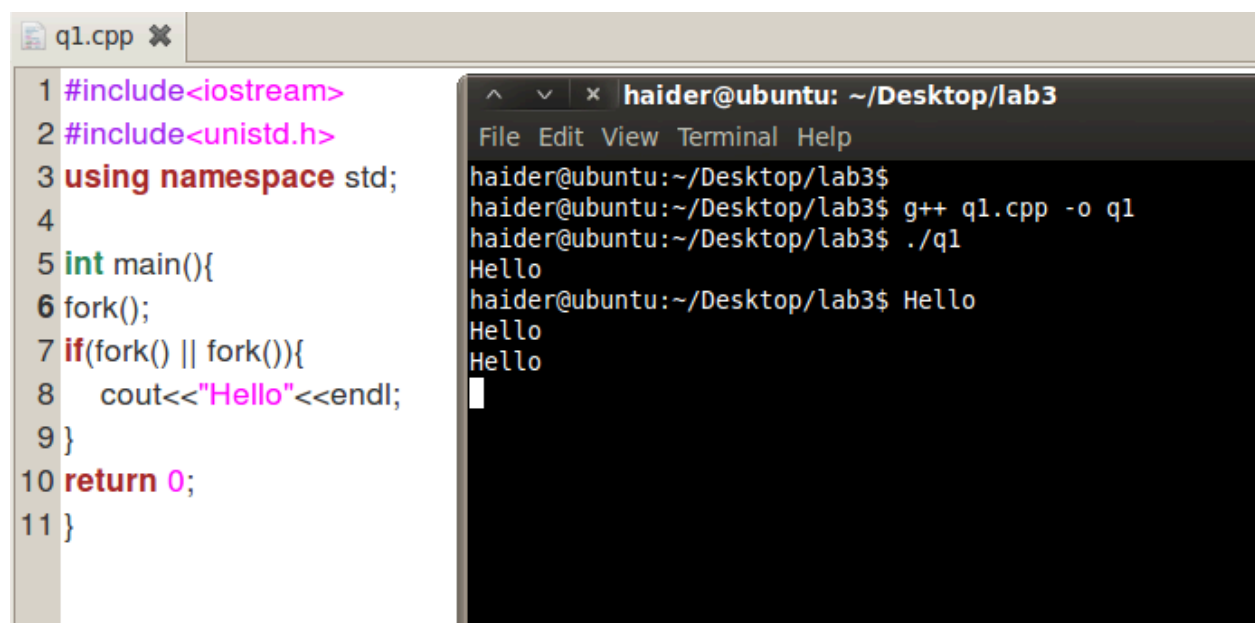
a)



```
1 #include<iostream>
2 #include<unistd.h>
3 using namespace std;
4
5 int main(){
6
7     if(fork() || fork()){
8         cout<<"Hello"<<endl;
9     }
10    return 0;
11 }
```

```
haider@ubuntu: ~/Desktop/lab3
File Edit View Terminal Help
haider@ubuntu:~/Desktop/lab3$ g++ q1.cpp -o q1
haider@ubuntu:~/Desktop/lab3$ ./q1
Hello
haider@ubuntu:~/Desktop/lab3$
```

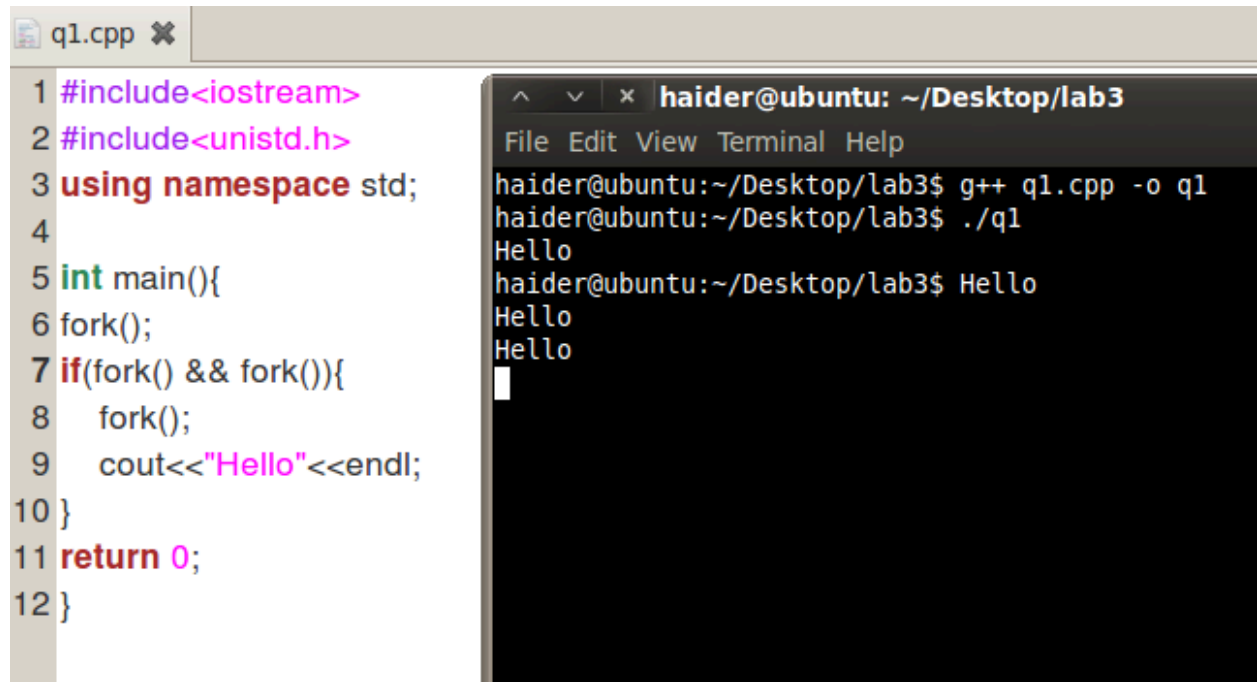
b)



```
1 #include<iostream>
2 #include<unistd.h>
3 using namespace std;
4
5 int main(){
6     fork();
7     if(fork() || fork()){
8         cout<<"Hello"<<endl;
9     }
10    return 0;
11 }
```

```
haider@ubuntu: ~/Desktop/lab3
File Edit View Terminal Help
haider@ubuntu:~/Desktop/lab3$
haider@ubuntu:~/Desktop/lab3$ g++ q1.cpp -o q1
haider@ubuntu:~/Desktop/lab3$ ./q1
Hello
haider@ubuntu:~/Desktop/lab3$ Hello
Hello
Hello
haider@ubuntu:~/Desktop/lab3$
```

c)



The image shows a C++ source file named `q1.cpp` and its execution output in a terminal window.

Source Code (q1.cpp):

```
1 #include<iostream>
2 #include<unistd.h>
3 using namespace std;
4
5 int main(){
6     fork();
7     if(fork() && fork()){
8         fork();
9         cout<<"Hello"<<endl;
10 }
11 return 0;
12 }
```

Terminal Output:

```
haider@ubuntu: ~/Desktop/lab3
File Edit View Terminal Help
haider@ubuntu:~/Desktop/lab3$ g++ q1.cpp -o q1
haider@ubuntu:~/Desktop/lab3$ ./q1
Hello
haider@ubuntu:~/Desktop/lab3$ Hello
Hello
Hello
```

The terminal output shows three lines of "Hello", indicating that the program successfully executed and produced the expected output.